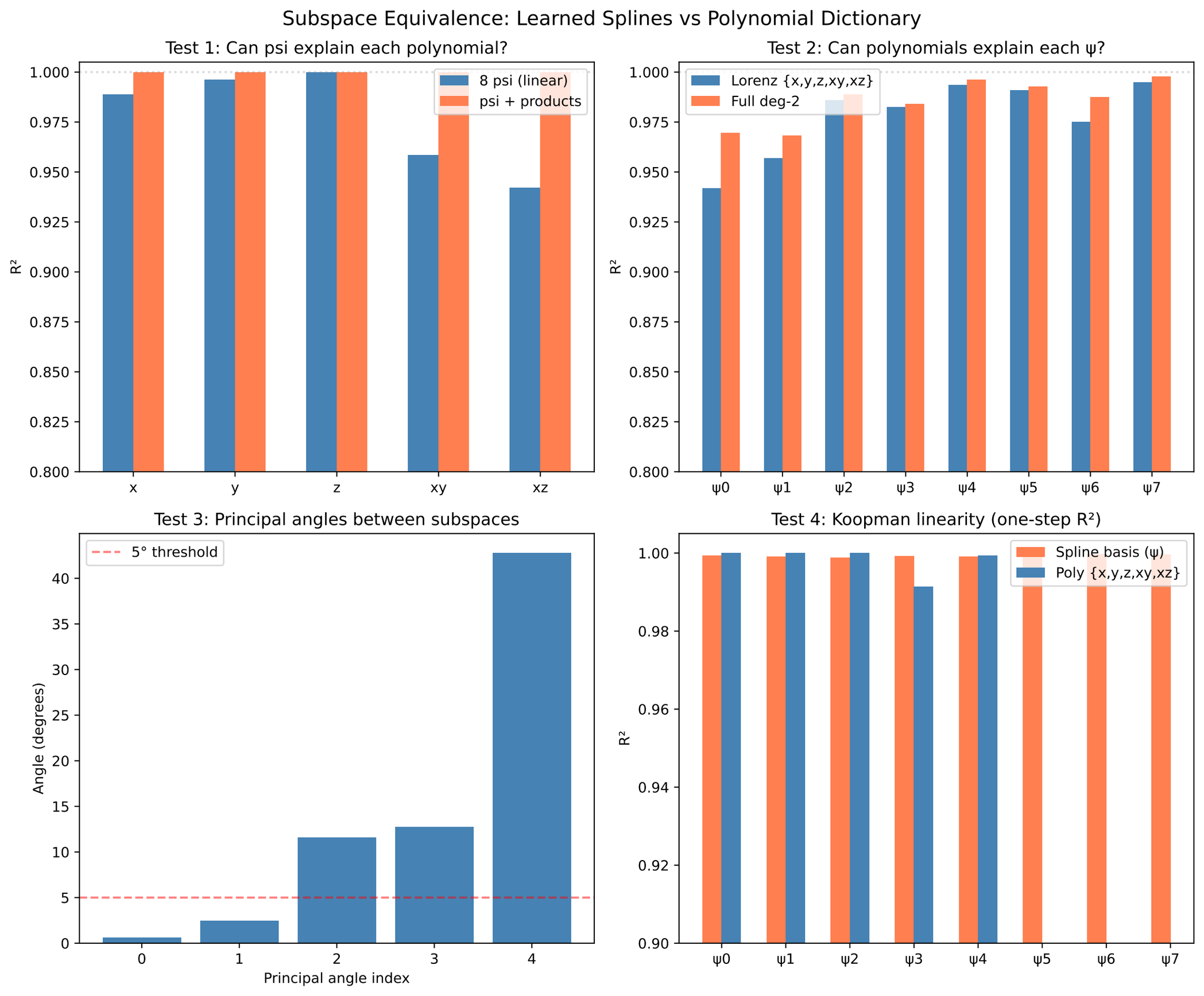
Subspace Equivalence: Learned Splines vs Polynomial Dictionary
Test 1: Can psi explain each polynomial?
8 psi (linear)
psi + products
R²
x
y
z
xy
xz
Test 2: Can polynomials explain each ψ?
Lorenz {x,y,z,xy,xz}
Full deg-2
ψ0
ψ1
ψ2
ψ3
ψ4
ψ5
ψ6
ψ7
Test 3: Principal angles between subspaces
5° threshold
Angle (degrees)
Principal angle index
Test 4: Koopman linearity (one-step R²)
Spline basis (ψ)
Poly {x,y,z,xy,xz}